

420-SN1-RE Programming in Science - Exercises for Topic 5 - Collections

These are practice questions *similar* to what you *might* see on a quiz or exam. Remember you will not have access to a computer during a test, so try to answer these without referring to Python. Once you've tried to answer the questions, feel free to check your answers with the Python environment.

A. Answer the following questions by deciding what the program would print.

1. What would the following program print?

```
x = "marianopolis"  
y = "college"  
print(x[7:9] + y[3:5] + x[:5:3] + y[0])
```

2. What would the following program print?

```
x = [100, 200, 300]  
y = [400, 500]  
print((x + y)[3], (y + x * 2)[7])
```

3. What would the following program print?

```
x = [1.0, 2.1, 2.0, 1.4, 1.1]  
t = 0  
k = 1  
while k < len(x):  
    t = t + x[k]  
    x[k] = x[k] + x[k-1]  
    k = k + 1  
print(t, x[4], k)
```

4. What would the following program print?

```
x = ["red", "blue", "yellow"]  
y = ["orange", "purple", "green"]  
z = x[2:] + x[:2] + y[1:]  
print(len(z), z[0], z[3])
```

5. What would the following program print?

```
x = "reviled"
y = ''
i = 0
while i < len(x):
    y = x[i] + y
    i = i + 1
print(y)
```

B. Write a program that meets the specifications given. Each of these problems was adapted from the Canadian Computing Competition.

1. Once upon a time, before emojis existed, people had to use simple “emoticons” created by 2 or 3 standard characters. For example, :-) is a “smiley face”, and :- (is a sad face.

Ask the user to input a string. Count the number of smiley faces and sad faces in the string. If the number of smiley faces is greater than the number of sad faces, print Happy. If the number of sad faces is greater than the number of sad faces, print Sad. If the numbers are equal, print Neutral.

For example, if the user typed:

Good morning! :-) Too bad it’s raining today :- (

Your program should print Neutral.

As a hint, consider writing a loop that would let you examine every three-character slice of the string to see if it matches either emoticon.

2. You supervise a small parking lot which has N parking spaces. Yesterday, you recorded which parking spaces were occupied by cars and which were empty. Today, you recorded the same information. How many of the parking spaces were occupied both yesterday and today? **Input Specification** The first line of input contains the integer N ($1 \leq N \leq 100$). The second and third lines of input contain N characters each. The second line of input records the information about yesterday’s parking spaces, and the third line of input records the information about today’s parking spaces. Each of these $2N$ characters will either be C to indicate an occupied space or . to indicate it was an empty parking space. **Output Specification** Output the number of parking spaces which were occupied both yesterday and today.

Sample Input 1

```
5
CC. .C
.CC. .
```

Output for Sample Input 1

```
1
```

Explanation of Output for Sample Input 1 Only the second parking space from the left was occupied both yesterday and today.

Sample Input 2

7
CCCCCCC
C.C.C.C

Output for Sample Input 2

4

Explanation of Output for Sample Input 2 The first, third, fifth, and seventh parking spaces were occupied yesterday and today.

3. Amy's parents were too cheap to buy her a smart phone, so they got her an old-fashioned flip phone. Instead of having a "proper" touchscreen keyboard, her phone just has a keypad like this:

1	2 abc	3 def
4 ghi	5 jkl	6 mno
7 pqrs	8 tuv	9 wxyz

Typing a text message requires patience, as in order to type a particular letter you need to press the associated numeric key between 1 and 4 times. For example, to enter a 'k' Amy has to press 5 twice, to produce an 's' she has to press 7 four times.

A problem arises with this scheme. If you want to type the word 'cab', you need to press 222 for the 'c', 2 for the 'a', and 22 for the 'b'. But if you just pressed 222222 in rapid succession, the phone might think you meant to type 'cc'. To avoid this ambiguity, you need to pause before attempting to type another letter that uses the same key, so the word 'cab' becomes 222-pause-2-pause-22. If the next letter is on a different key, there's no need to pause.

Amy is trying to lobby her parents for a better phone, so she decides to write a program that will calculate how long it will take her to type a particular message. She assumes it takes one second to press a key, and two seconds for a pause, so the word 'cab' will require 10 seconds.

Write a program to compute how long many seconds it takes to type a series of words. Each word will be entered in lower-case letters on a separate line, and program will exit when the line is blank (empty). After reading each word, your program should print the number of seconds required to type that word. A session with your program should look like this (the input is in red, the program's response is in black).

```
Enter a word: a
1
Enter a word: map
3
Enter a word: feed
14
Enter a word: cell
13
```

Note: this problem is pretty tricky. Be sure to ask for a hint if you get stuck.